WEEWARE | PALAWAN

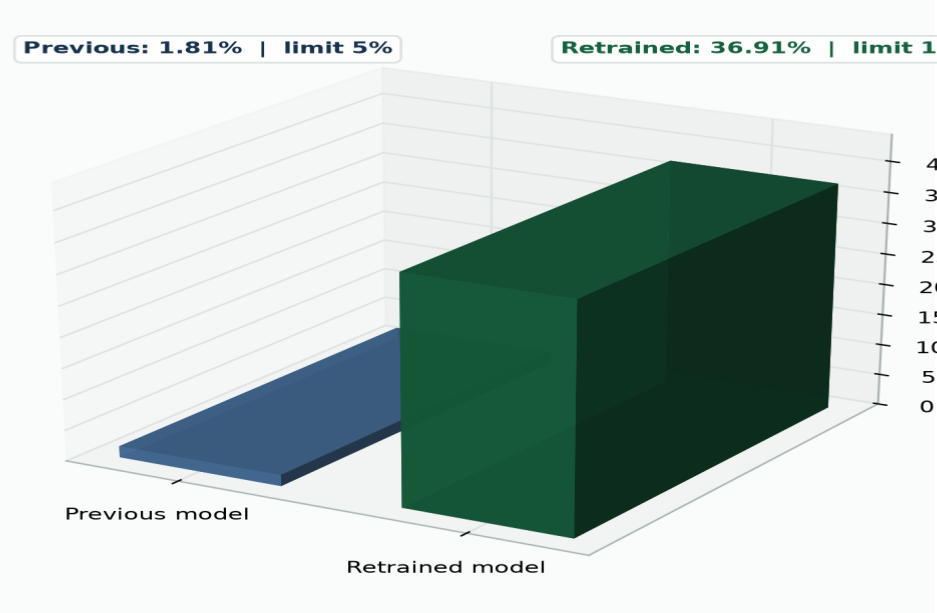
How the new Palawan forecast was trained

A plain-language guide to the data, checks, and old-versus-new result

Prepared 10 November 2026 | Training data: 2014-2025 | Example forecast: 2026

100	23	12	5
training entries	labeled places	years covered	weather measures

AFTER TRAINING | same 2026 weather information



2026-01-01 to 2026-12-31 | Puerto Princesa | new minus old = +35.10 percentage points

The old model limit was 5%; the new model limit is 100%. The source infestation numbers are not fully comparable.

The new model is trained and connected to the local WeeWare website. Important: its much higher estimate does not prove that real infestation increased or that the new model is more accurate. The two models learned from different kinds of infestation records.

The short version

We gave both models the same weather information for Puerto Princesa during 2026. The older model estimated 1.81%; the retrained model estimated 36.91%. The new estimate is 35.10 percentage points higher.

That gap is large, but the old model was trained on values between 0 and 5%, while the new training file includes values as high as 52.15%. The old model also cannot report above 5%. The input records are not fully comparable, so this test shows how the two formulas differ; it does not tell us which answer is closer to the true infestation rate.

What the checks found

When we hid one whole year at a time, the new model's average miss was 15.00 points. A simple guess using the average training value missed by 17.73 points. That is an improvement on this check.

When we hid one whole place at a time, the new model's average miss was 17.76 points versus 18.59 for the simple guess. But its score that penalizes large misses was worse: 20.41 versus 19.46. This does not yet show dependable performance in a place absent from training.

Many places have the same supplied infestation value repeated in several years. So the 100 entries do not represent 100 fully independent, year-specific field measurements. This may make the year check look better than it would on new observations.

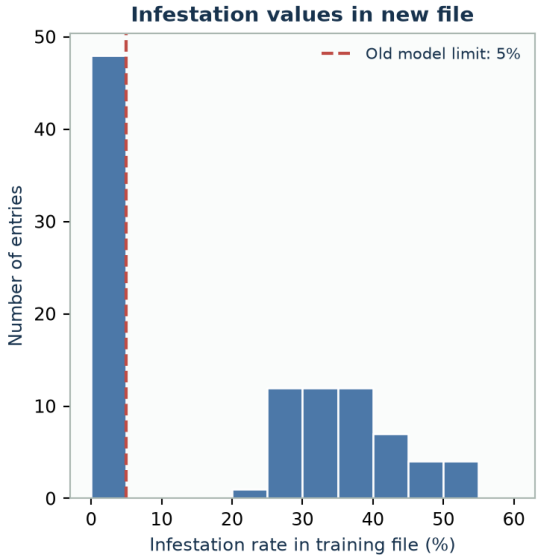
What happened

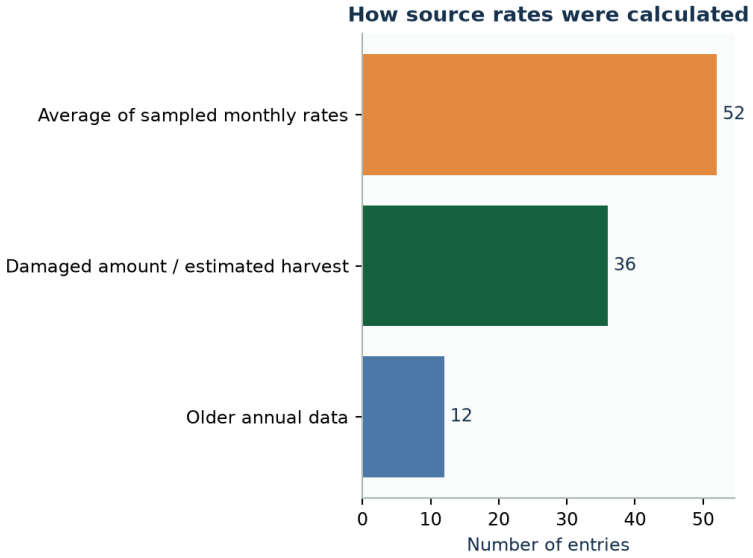
Step	In everyday terms
Before	The website used an older model trained on 12 annual Puerto Princesa records. It was limited to estimates from 0% to 5%.
Prepare	Annual weather information was assembled for Palawan places. Existing climate-model values were compared with NASA weather estimates where both were available.
Train	The model was fitted to 100 place-and-year entries with five weather measures. Checks hid whole years and whole places to see how estimates held up.
After	The updated model was made active in the website. Both saved models were given the same 2026 weather information for a direct output comparison.
Website	The local website, data service, and model service responded during the check. The active model reported 100 training entries and 23 labeled places.

LASSO is the name of the math method used here. It learns a formula that combines the five weather measures. It is not a neural network, so it does not have 'epochs'. This report shows the checks and prediction evidence instead of an epoch chart.

What went into training

Item	What the file contains
Training entries	100 rows from 23 labeled places and 12 years, from 2014 to .
Weather	Rainfall, temperature, humidity, sunlight, and wind. All 100 entries have values for these five measures.
Infestation values	From 0.00% to 52.15%. The middle value is 26.00%. 36 entries are recorded as zero.
52 entries	An average of sampled monthly infestation rates. The source did not include harvest amounts for this calculation.
36 entries	Total damaged volume divided by estimated harvest, from the supplied workbooks.
12 entries	Values retained from the older Puerto Princesa annual dataset.
Repeated values	65 entries repeat a place's reported rate in another year. The typical place has only 1 distinct supplied target value(s).





Why repeated values matter

A place can have different weather in 2014, 2017, 2021, and 2025. But if its infestation number is copied unchanged across those years, the model is not learning how infestation changed year to year at that place. We need a truly measured value for each year to check whether changing weather predicts changing infestation.

Where the weather came from

Historical weather was taken from NASA POWER monthly estimates and combined into annual values. Coordinates are town or municipality centers, not farm locations. They support area-level estimates, not weather readings from a specific orchard. Puerto Princesa is included; Narra has weather data but no infestation label in the supplied files, so it could not be used as a labeled training place.

How the climate sources compared

CMIP6 is the name of the global climate-model collection used here. Its local model files were compared with estimates from NASA POWER, a NASA service that provides weather and sunlight estimates for a location for months covered by both. A positive difference means the climate-model value was higher on average. The 'usual gap' is the typical size of the difference. This describes agreement between gridded sources; it is not proof of exact weather at any farm.

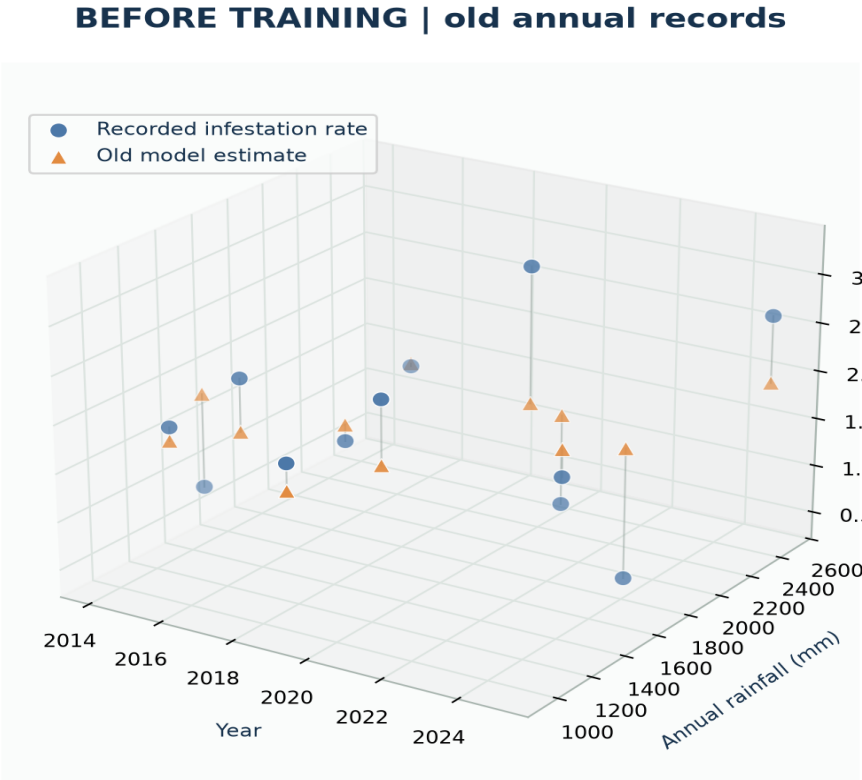
Weather and years	Model average vs NASA	Usual gap
Rainfall, 2015-2020	+48.37 mm/month	102.25 mm/month
Rainfall, 2021-2025	+38.36 mm/month	101.72 mm/month
Temperature, 2023-2025	+1.20 °C	1.23 °C
Humidity, 2023-2025	+8.14 percentage points	9.15 percentage points
Wind, 2025-2025	-0.54 m/s	0.97 m/s
Solar, 2014-2014	-0.14 MJ/m²/day	2.37 MJ/m²/day

For rainfall in 2021-2025, climate-model values averaged about 38 mm/month higher than NASA POWER, and a typical month differed by about 102 mm. The 2026 example below used climate-model values for rain, temperature, humidity, and wind. Solar radiation was missing from the local 2026 files, so the service used an average from the training data instead.

In the older Puerto Princesa weather table, the average difference from NASA POWER was 173 mm/year for rain and 0.67 degrees C for temperature. The differences were smaller for humidity, sunlight, and wind. Historical training weather was therefore prepared using one consistent NASA POWER source.

Before training: the older model

Each blue circle is a recorded infestation value from the old annual dataset. Each orange triangle is the old model's estimate for the same record. The line between them shows the gap, or miss. The chart also shows the year and annual rainfall.

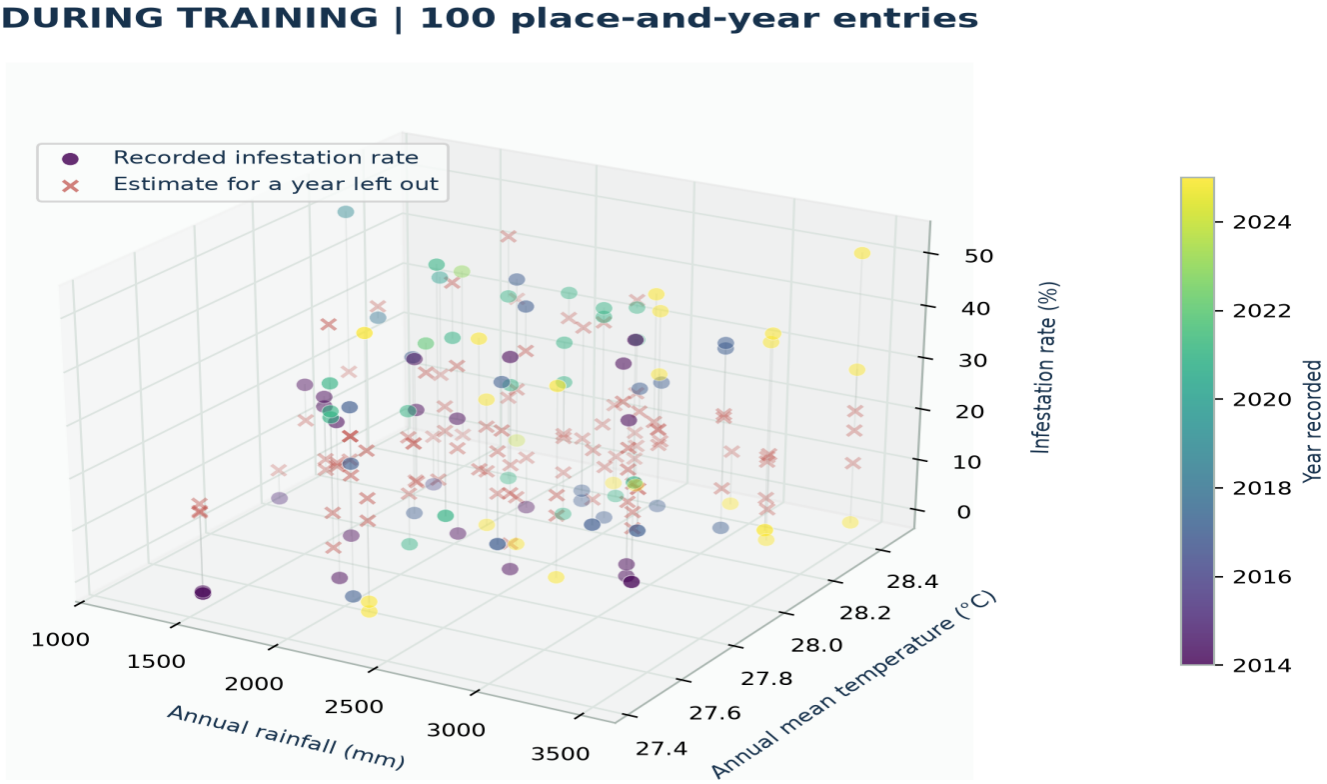


Each blue circle or orange triangle is one year. The connecting line shows how far the estimate was from the recorded rate.

The old model used 12 Puerto Princesa annual records and could not report above 5%.

During training: checking the new data

Each mark represents one place in one year. The graph compares rain and temperature with the supplied infestation value. Red x marks show estimates for a year hidden while the model learned; all places from that year were hidden together.

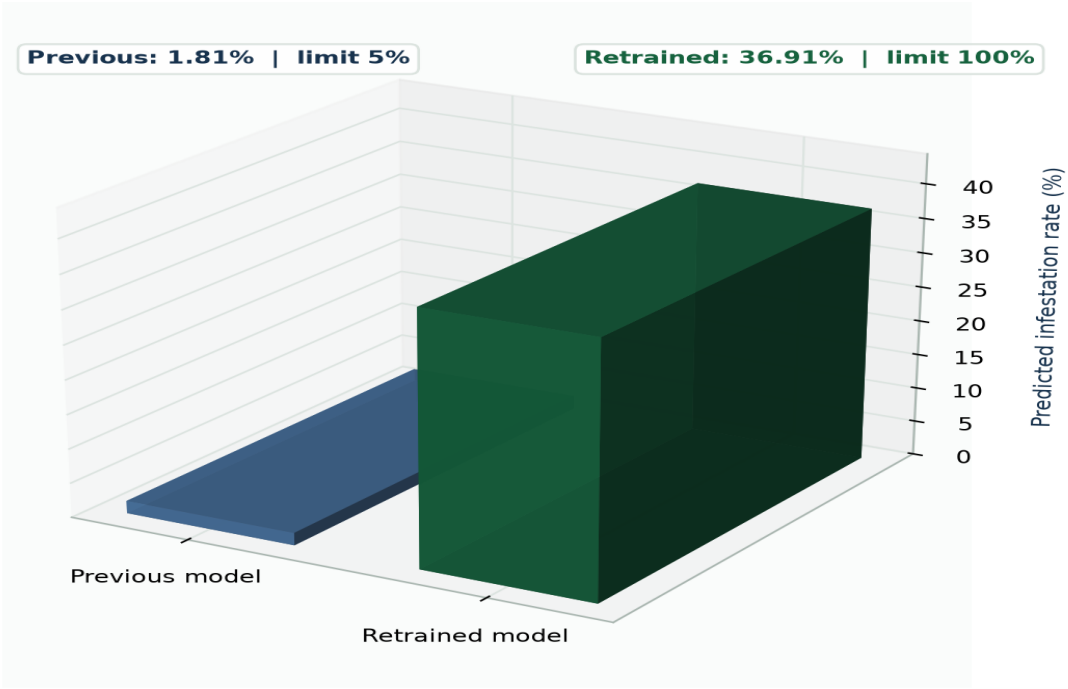


For each red x estimate, the model learned without seeing any entries from that year.

This is a picture made from saved training and check results, not a camera image of training as it ran. The model learns a formula; it does not produce epoch-by-epoch progress pictures.

After training: one weather test, two models

We gave both models the same weather information for January 1 to December 31, 2026, at the Puerto Princesa reference coordinate (9.7494 north, 118.7384 east). The bars show what each model estimated.



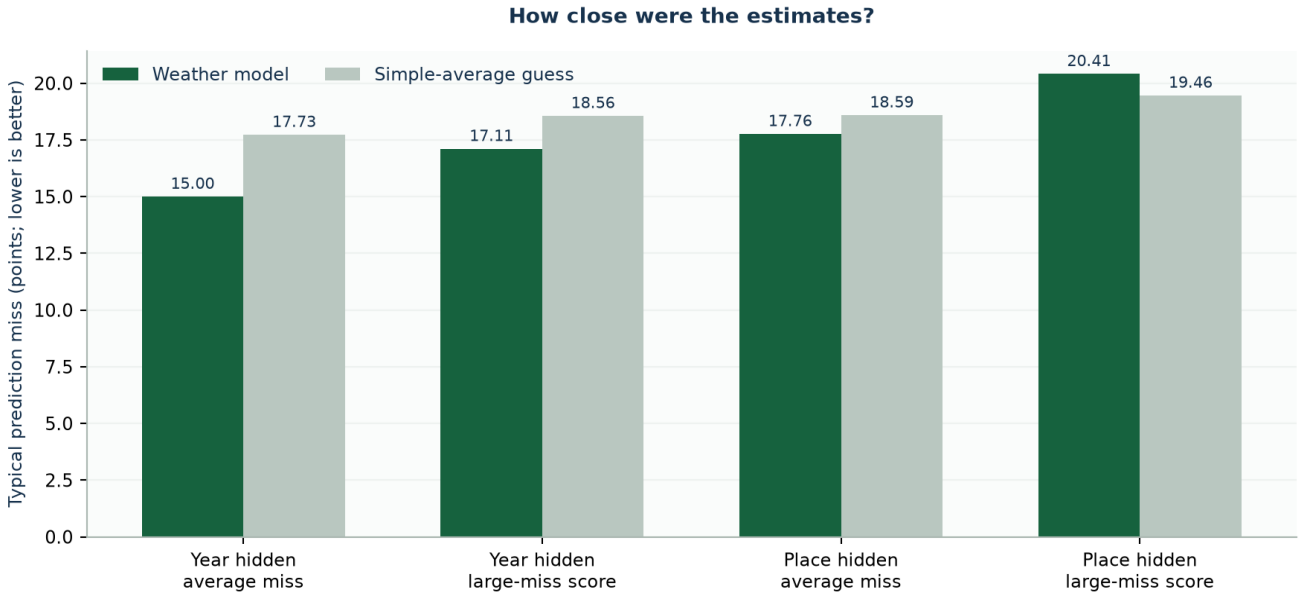
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The estimates are not a fair accuracy contest: the models used different infestation records and different maximum output values. This comparison shows how the model result changed, not which result is correct.

How to read the model checks

Think of validation as a practice test: hide some records, let the model learn from the rest, then compare its guesses with the hidden records. We also compare it with a simple guess that always uses the average infestation value from the visible records.



Year check: the model's average miss was 15.00 percentage points, compared with 17.73 for the simple guess. Its score that penalizes especially large misses was 17.11, compared with 18.56. Lower numbers are better.

Place check: average miss was 17.76 for the model and 18.59 for the simple guess. But the large-miss score was 20.41 for the model and 19.46 for the simple guess. This is mixed evidence, not proof the model works reliably in a new municipality.

Average miss is the typical distance between a guess and the supplied value. The large-miss score gives especially bad guesses more penalty. The unit is percentage points: for example, the gap between 20% and 15% is 5 percentage points. Repeated infestation values across years may make the year check look better than it would on new field observations.

The 2026 result in detail

Model	Estimate	Maximum allowed	Risk label and cutoffs
Previous	1.81%	5%	Medium (middle band: 1.5%-3%)
Retrained	36.91%	100%	Medium (middle band: 33%-66%)

Weather information used

Measure	Value	Source
Rainfall	2574.57 mm/year	Climate model
Temperature	29.64 degrees C	Climate model
Relative humidity	88.34 %	Climate model
Solar radiation	18.14 sunlight per day	Training-data average (fallback)
Wind speed	4.64 m/s	Climate model

Both models say 'Medium', but that label uses different cutoffs in each model. It is a software category, not a verified biological danger level. The estimate is not a measured farm infestation rate, and the app does not provide a confidence range.

What we can and cannot conclude

What we know

The updated model was trained, saved as the active model, and returned an estimate through the WeeWare prediction service. It uses five weather measures and records for multiple Palawan places. Its year-check scores beat the simple average guess in this dataset.

What we do not know yet

- Whether 36.91% is closer to reality than 1.81%; the source records and output limits differ.

- Whether the model works reliably in every Palawan municipality; the place check was mixed.

- Whether year-to-year weather changes predict year-to-year infestation; many supplied values repeat across years.

- The exact weather at a farm; the historical coordinates are town centers and climate grids cover broad areas.

Best next step

Confirm that all infestation numbers use the same definition. Replace repeated values with actual year-specific observations where possible. Then compare future predictions with new orchard survey results and keep the simple-average guess as a benchmark.

A few words explained

Term	Plain meaning
Model	A formula fitted to past examples that estimates a result from new inputs.
Training entry	One row pairing one place, one year, weather values, and an infestation value.
Validation / practice test	Checking guesses on records kept out while the formula is fitted.
LASSO	The kind of formula used. It combines weather measures while limiting overly complicated weights.
CMIP6	A collection of computer models used to estimate possible future climate.
NASA POWER	A NASA service that provides weather and sunlight estimates for a location.
Alpha	A setting that controls how strongly the formula is kept simple. This run used 0.05.
Average miss	The average distance between a guess and the supplied value; lower is better.
Large-miss score	An error score that penalizes especially large misses more; lower is better.
Percentage point	The direct difference between two percentages. 20% minus 15% is 5 points.
Fallback	A substitute value used when a weather measure is missing from local climate files.

Sources and saved records

Weather: NASA POWER monthly API guide. Place list: Philippine Statistics Authority - Palawan PSGC page. Climate model catalog: CMIP6 CanESM5. Coordinates: OpenStreetMap Nominatim town-center lookups.

Training record: 100 entries from 2014-2025, experiment 35a8782a52494c5ea97dc3f086018427. Saved files include the prepared training table, the detailed source table, the experiment record, and the active model. The final model used five weather measures, a setting called alpha of 0.05, and up to 100,000 solver steps. These technical settings are included for anyone who needs to reproduce the work.